

Corrective RAG, Self-Evaluation, and Feedback Substitution

What Happens When an AI System Grades Its Own Grounding?

AI Memory and Reality Drift - Note #6 v1

A. Jacobs - Semantic Fidelity Project

The Basic Pattern

Corrective and self-reflective RAG systems attempt to improve reliability by evaluating retrieved evidence before or during generation. A model may grade passage relevance, decide whether more retrieval is needed, critique its answer, or determine whether a claim is adequately supported.

These mechanisms add an important feedback loop. The system no longer accepts every retrieval result blindly. But the feedback often comes from another language-model judgment operating over the same representations and sharing similar limitations.

The system gains correction behavior without necessarily gaining new contact with the external world.

When Evaluation Replaces Verification

A retrieval grader can identify obviously irrelevant passages. A critique prompt can detect unsupported claims. A verifier model can compare an answer with supplied context. These functions are useful because they make quality control scalable.

The structural risk appears when internal agreement is treated as external validation. A model can judge a passage relevant because it matches the same mistaken framing used to retrieve it. It can judge an answer faithful because the context and answer share a compressed error.

How Feedback Substitution Emerges

Stage 1 - Retrieval. The system selects evidence from an indexed representation.

Stage 2 - Evaluation. A model scores relevance, support, or groundedness.

Stage 3 - Adaptation. The system retrieves again, rewrites, or revises its answer according to the score.

Stage 4 - Internal closure. Improvement is measured through another model-based representation of quality.

Stage 5 - Feedback substitution. Internal coherence takes the operational place of independent correction.

Feedback Substitution and Reality Drift

Feedback Substitution occurs when a system replaces feedback from outcomes, sources, or external observers with feedback generated inside its own representational environment. The loop can become more sophisticated while remaining closed.

Reality Drift describes the larger condition. The system continues correcting itself, metrics improve, and outputs become more polished, yet the corrective process may become progressively less answerable to anything outside the model-index-evaluator stack.

The Representation Stack

Source becomes indexed evidence. Evidence becomes retrieved context. Context becomes answer. Answer and context become inputs to an evaluator. The evaluator's score becomes the signal that reorganizes retrieval and generation.

If every layer descends from the same incomplete representation, agreement across layers can look like independent confirmation. A reality-constrained system needs heterogeneous checks: source reopening, deterministic validation, external tools, outcome tests, or human review where stakes justify it.

Examples Across AI Systems

Self-RAG: The model decides whether a response needs retrieval and whether retrieved passages support its claims.

Corrective RAG: A relevance grader triggers alternative search when the initial evidence is judged weak.

Agent workflows: A critic model approves a plan created by another model using the same memory store.

Evaluation pipelines: LLM judges score groundedness without direct access to the domain beyond the supplied context.

Conclusion

Self-evaluation can make RAG systems more adaptive and reliable. It should be understood as an additional representation of quality, not as proof of contact.

The critical design question is where genuinely independent correction enters the loop. When every correction is produced inside the same architecture, the system can become better at agreeing with itself.

Related Phrases and Concepts

- corrective RAG
- CRAG
- self-RAG
- retrieval grading
- adaptive retrieval
- self-reflection
- verifier models
- feedback substitution

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)